

Review of Chapter 3.

§ 3.1.

1. General form of second-order linear differential equation.

$$A(x)y'' + B(x)y' + C(x)y = F(x).$$

or

$$y''(x) + P(x)y' + Q(x)y = f(x) \quad (\text{if } A(x) \neq 0)$$

i) $f(x) \equiv 0 \rightarrow$ homogeneous equation

ii) $f(x) \not\equiv 0 \rightarrow$ nonhomogeneous equation.

2. Let f, g be two functions defined on I .

Wronskian $W(f, g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = fg' - f'g$

Let y_1, y_2 be two solutions of $y'' + P(x)y' + Q(x)y = 0$

i) y_1, y_2 linearly dependent $\Leftrightarrow W(y_1, y_2) \equiv 0$ on I .

ii) y_1, y_2 linearly independent $\Leftrightarrow W(y_1, y_2) \neq 0$ on every point of I .

3. General solution of homogeneous equation

$$y'' + P(x)y' + Q(x)y = 0$$

is given by

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

where y_1, y_2 are any two linear independent solutions of

$$y'' + P(x)y' + Q(x)y = 0$$

4. The case of constant coefficients

$$ay'' + by' + cy = 0$$

where a, b, c are constants.

Characteristic equation: $ar^2 + br + c = 0$

let r_1, r_2 be two roots of the characteristic equation.

i) $r_1 \neq r_2$, real, then

$$y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

ii) $r_1 = r_2 = r$, real, then

$$y(x) = (C_1 + C_2 x) e^{rx}$$

iii) r_1, r_2 are complex conjugate pairs: $r_1 = a + bi, r_2 = a - bi$, then

$$y(x) = e^{ax} (C_1 \cos bx + C_2 \sin bx).$$

§3.2.

1. General n th-order linear differential equation.

$$P_0(x) y^{(n)} + P_1(x) y^{(n-1)} + \dots + P_{n-1}(x) y' + P_n(x) y = F(x)$$

i) $f(x) \equiv 0 \rightarrow$ homogeneous equation

ii) $f(x) \not\equiv 0 \rightarrow$ nonhomogeneous equation.

$$\rightarrow y^{(n)} + P_1(x) y^{(n-1)} + \dots + P_{n-1}(x) y' + P_n(x) y = f(x)$$

Initial value problem

$$\begin{cases} y^{(n)} + P_1(x) y^{(n-1)} + \dots + P_{n-1}(x) y' + P_n(x) y = f(x) \\ y(a) = b_0, y'(a) = b_1, \dots, y^{(n-1)}(a) = b_{n-1} \end{cases} \Rightarrow \text{unique solution.}$$

2. Wronskian

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix} \leftarrow \text{determinant of an } n \times n \text{ matrix.}$$

Let $y_1, y_2, \dots, y_n$ be n solutions of the homogeneous equation, then

i) $y_1, y_2, \dots, y_n$ linearly independent $\Leftrightarrow W(y_1, \dots, y_n) \neq 0$ on I .

ii) $y_1, y_2, \dots, y_n$ linearly dependent $\Leftrightarrow W(y_1, \dots, y_n) \equiv 0$ on I .

3. General solution of $y^{(n)} + P_1(x)y^{(n-1)} + \dots + P_{n-1}(x)y' + P_n y = 0$ is given by

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + \dots + C_n y_n(x)$$

where $y_1(x), y_2(x), \dots, y_n(x)$ are n linearly independent solution of the homogeneous equation.

§ 3.3.

Consider

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_2 y'' + a_1 y' + a_0 y = 0 \rightarrow \text{homogeneous equation}$$

where $\{a_i\}_{i=0}^n$ are real constants.

Characteristic equation:

$$a_n r^n + a_{n-1} r^{n-1} + \dots + a_2 r^2 + a_1 r + a_0 = 0.$$

roots: $r_1, r_2, \dots, r_n$

i) distinct roots

$$y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x} + \dots + C_n e^{r_n x}$$

ii) Repeated roots.

root: $\underline{r}$ with multiplicity $\underline{k}$. then the part of general

solution corresponding to r is of the form

$$(C_1 + C_2 x + C_3 x^2 + \dots + C_k x^{k-1}) e^{rx}$$

iii) Complex root

unrepeated pair of complex conjugate roots $r = a \pm bi$. the

the part of the general solution corresponding to $a \pm bi$

is of the form

$$e^{ax} (C_1 \cos bx + C_2 \sin bx).$$

§ 3.4.

1. A mass-spring-dashpot system

$$mX'' = F_S + F_R + F_E \quad \rightarrow \text{external force.}$$

$\downarrow$ Restorative force $\rightarrow$ force by dashpot

$$F_S = -kX \quad (\text{Hooke's law}).$$

$$F_R = -cV = -cX'$$

$$F_E = f(t)$$

$$\Rightarrow mX'' = -kX - cX' + f(t)$$

$$mX'' + cX' + kX = f(t)$$

$$c=0 \rightarrow \text{undamped motion}$$

$$c>0 \rightarrow \text{damped motion.}$$

$$f(t) \equiv 0 \rightarrow \text{free motion}$$

$$f(t) \not\equiv 0 \rightarrow \text{forced motion.}$$

3. Free undamped motion.

$$mX'' + kX = 0$$

$$\Rightarrow X(t) = A \cos \omega_0 t + B \sin \omega_0 t \quad \text{where } \omega_0 = \sqrt{\frac{k}{m}}$$

$$= C \cos(\omega_0 t - \alpha)$$

where 

$$C = \sqrt{A^2 + B^2}$$

$$\cos \alpha = \frac{A}{C}, \quad \sin \alpha = \frac{B}{C}$$

$$\tan \alpha = \frac{B}{A}.$$

$C \rightarrow$ Amplitude.

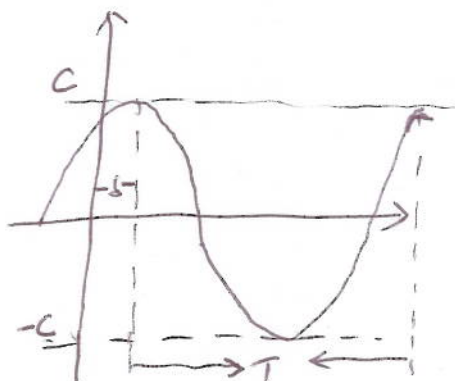
$\omega_0 \rightarrow$ Circular frequency.

$\alpha \rightarrow$ Phase angle.

$$\text{Period: } T = \frac{2\pi}{\omega_0}$$

$$\text{frequency: } \nu = \frac{1}{T} = \frac{\omega_0}{2\pi}$$

$$\text{time lag: } \delta = \frac{\alpha}{\omega_0}$$



§ 3.5.

1. Nonhomogeneous equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = f(x) \quad (Ly = f)$$

$$y = y_c + y_p$$

y_c - complementary solution or say general solution of the associated homogeneous equation $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$ ($Ly = 0$)

y_p - a particular solution for the nonhomogeneous equation.

2. Method of Undetermined Coefficient.

$f(x)$ is a linear combination of finite products of functions of the following three types.

1. A polynomial in x .
2. An exponential function e^{rx}
3. $\cos kx$ or $\sin kx$.

i). Let f be a sum of terms each of the form

$$P_m(x)e^{rx}\cos kx \text{ or } P_m(x)e^{rx}\sin kx. \quad (*)$$

$\hookrightarrow$ a polynomial in x of degree m .

then take as the trial solution

$$y_p(x) = x^s [(A_0 + A_1 x + A_2 x^2 + \dots + A_m x^m) e^{rx} \cos kx + (B_0 + B_1 x + B_2 x^2 + \dots + B_m x^m) e^{rx} \sin kx]$$

where s is the smallest nonnegative integer such that no term in y_p duplicates a term in the complementary solution y_c .
Finally, determine the coefficients by substituting y_p into the nonhomogeneous equation.

ii) It is common to have.

$$f(x) = \underline{f_1(x)} + \underline{f_2(x)}.$$

have the form of $\otimes$.

then $y_p(x) = y_p^1(x) + y_p^2(x)$

Typical examples

$f(x)$

$$P_m = b_0 + b_1x + b_2x^2 + \dots + b_mx^m$$

$$a \cos kx + b \sin kx$$

$$e^{rx}(a \cos kx + b \sin kx)$$

$$P_m(x)e^{rx}$$

$$P_m(x)(a \cos kx + b \sin kx)$$

y_p

$$x^s(A_0 + A_1x + A_2x^2 + \dots + A_mx^m)$$

$$x^s(A \cos kx + B \sin kx)$$

$$x^s e^{rx}(A \cos kx + B \sin kx)$$

$$x^s(A_0 + A_1x + A_2x^2 + \dots + A_mx^m)e^{rx}$$

$$x^s[(A_0 + A_1x + \dots + A_mx^m)\cos kx + (B_0 + B_1x + \dots + B_mx^m)\sin kx]$$